STEM Lesson Plan: Sound Waves & Wave Machines

Hands-On Exploration for Grade 3 Physical Science

Grade Level: 3rd Grade
Duration: 60 Minutes
Subject: STEM / Physical Science
NGSS Standard: 3-PS2-1, 3-PS2-3 (Wave properties / Vibrations)

1. Lesson Overview

In this interactive STEM lesson, students will discover how sound is created by vibrations and how those vibrations travel through mediums as waves. Students will build a hands-on "Duct Tape & Jelly Bean Wave Machine" to visualize the transverse motion of waves, helping bridge the gap between abstract sound concepts and concrete physical demonstrations.

2. Learning Objectives

- **Identify** that sound is produced by physical vibrations.
- **Explain** how energy travels through a medium via waves without the actual matter permanently moving spots.
- **Construct** a functional mechanical model (a wave machine) to observe and describe frequency, amplitude, and wavelength.

3. Key Vocabulary

Word	Definition for 3rd Graders
Vibration	A rapid back-and-forth movement that creates sound.
Sound Wave	An invisible pattern of energy that moves through the air, water, or objects.
Medium	The material (like solid, liquid, or gas) that a wave travels through.
Amplitude	The height of a wave, which determines how loud a sound is.
Wavelength	The distance from the top of one wave to the top of the very next wave.

4. Materials Needed (Per Student Team)

- 1 roll of heavy-duty duct tape (approx. 2-3 feet per machine)

- 30-40 wooden skewer sticks or popsicle sticks
- 1 pack of jelly beans, gummy candies, or clay balls (to attach to stick ends)
- 2 heavy books or desk clamps to anchor the ends of the tape
- A tuning fork and a small bowl of water (for teacher demonstration)

> **Teacher Tip: STEM Focus**
> Before launching the build, activate prior knowledge by asking students to touch their throats softly and say "Ahhh". Ask: *What do you feel?* Connect that physical sensation directly to the word **vibration**!

5. Guided Activity: Building the Wave Machine

Follow these structural steps to build and use the mechanical wave model:

Step 1 **Set up the Spine:** Stretch out a 2-foot piece of duct tape, sticky-side up, between two desks. Anchor both ends firmly with heavy books or clamps so the tape is taut.

Step 2 **Assemble the Weights:** Push a jelly bean onto both ends of every wooden skewer stick. Make sure they are pushed on equally to ensure balance.

Step 3 **Place the Sticks:** Lay the center of each skewer horizontally across the sticky tape, spacing them evenly about 1 inch apart along the entire length.

Step 4 **Secure the Line:** Place another layer of duct tape directly over the first piece, sandwiching the sticks securely in the middle.

Step 5 **Launch a Wave:** Gently tap the stick at one end of the machine. Watch as a wave of energy moves smoothly down the line to the other side!

6. Discussion & Concept Check

Have students watch closely as the wave travels down the machine. Ask them to notice if individual jelly beans move to the end of the tape, or if they stay in place while only the energy pulse travels.

Explain that the wave pattern follows basic wave rules where the relation of speed, frequency, and wavelength can be modeled simply as: $\textit{Wave Energy} = \textit{Frequency} \times \textit{Amplitude}$. When they tap the stick harder, the amplitude grows, representing a louder sound!

7. Assessment & Wrap-Up

1. What happens when you tap the wave machine harder vs. softer? (Connects to amplitude/volume).
2. Does the tape itself travel down the room? Why or why not? (Connects to energy transport vs. matter movement).
3. Draw your wave machine on an index card and label one peak (crest) and one valley (trough).